



A perspective on Cyber Security

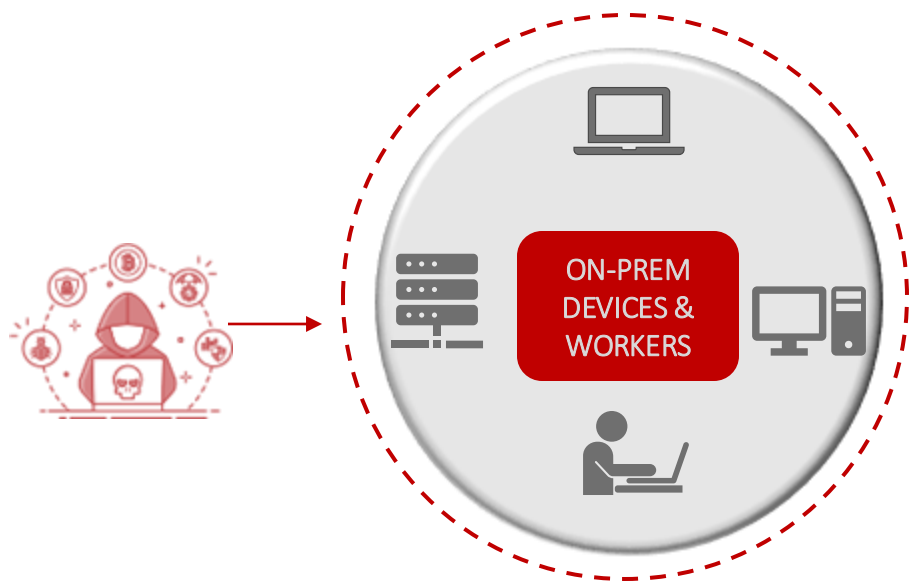
Lalit Mohan Sanagavarapu
Lalit.mohan@quickheal.com



Disappearing IT Boundaries

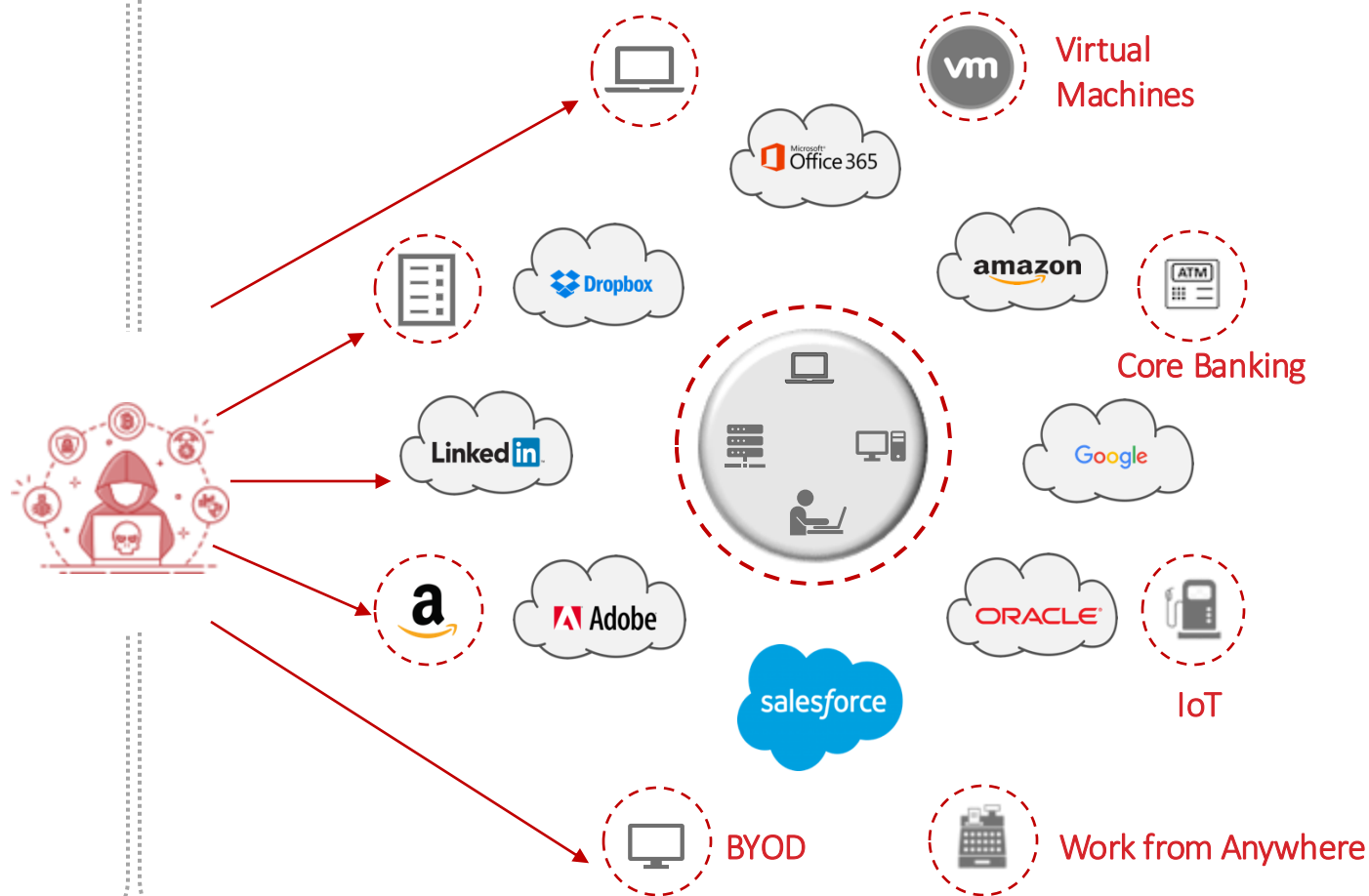
Before

The Network was the Perimeter



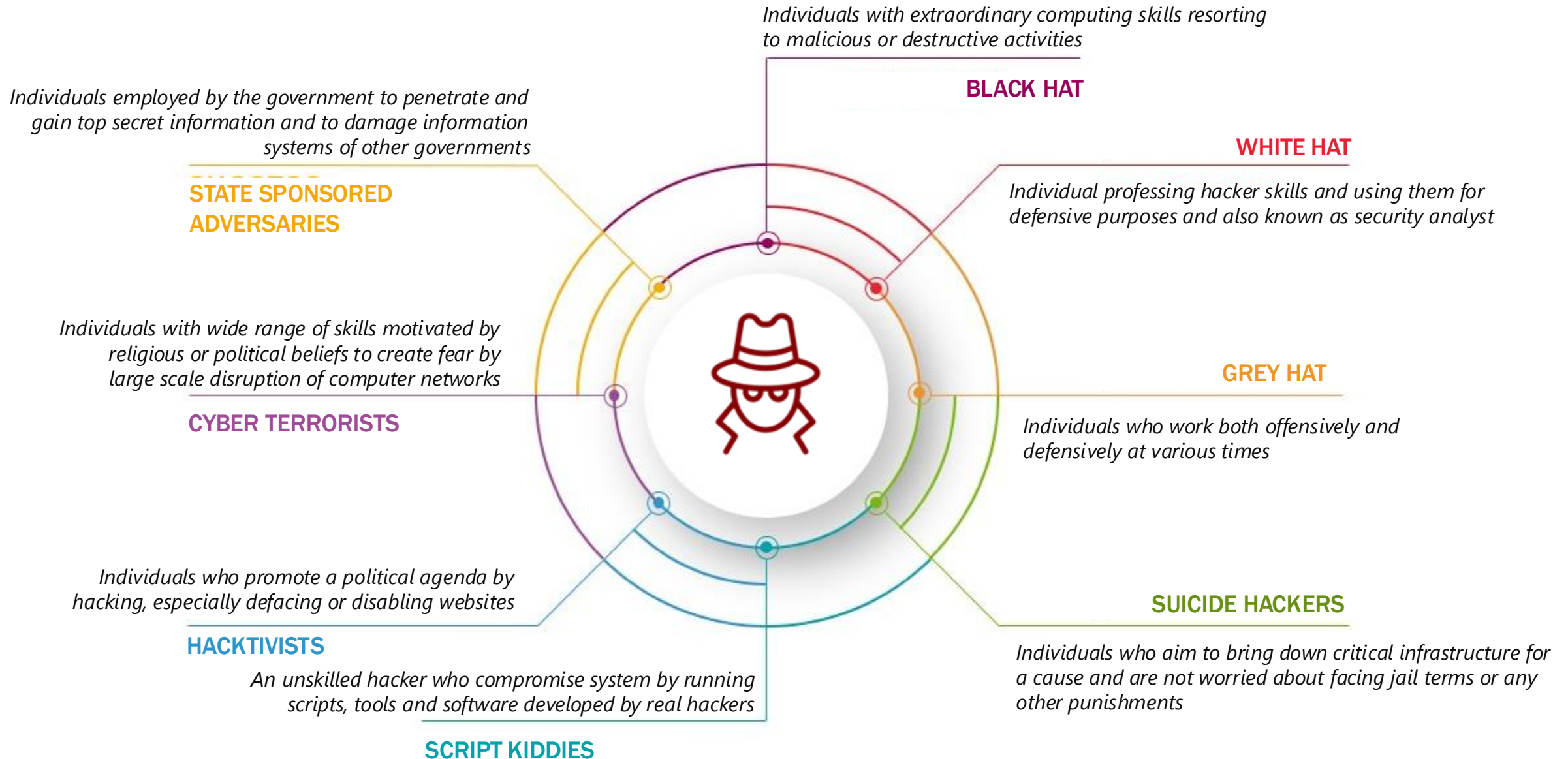
Today

The Endpoint is the NEW Perimeter



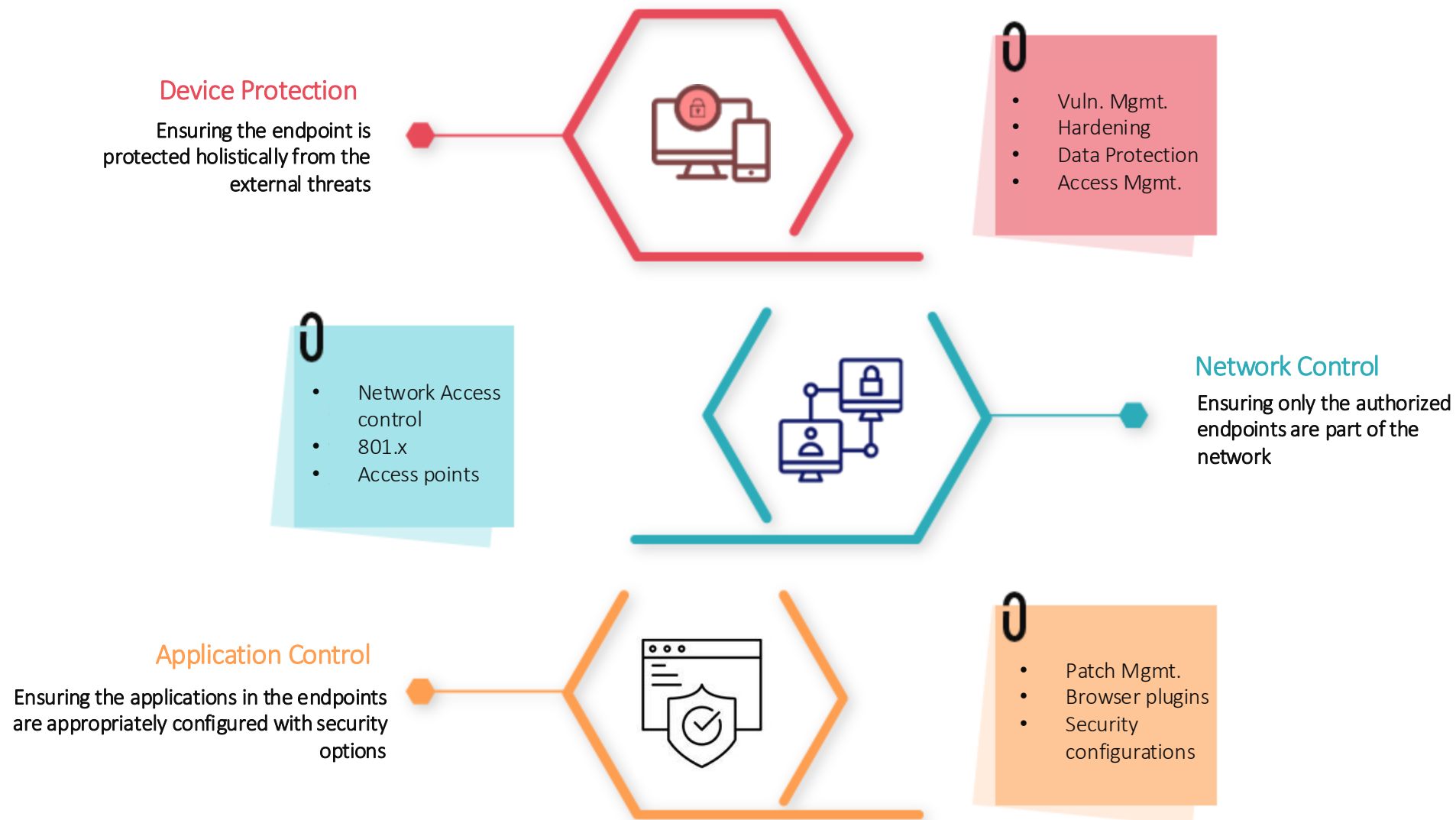


Threat Actors



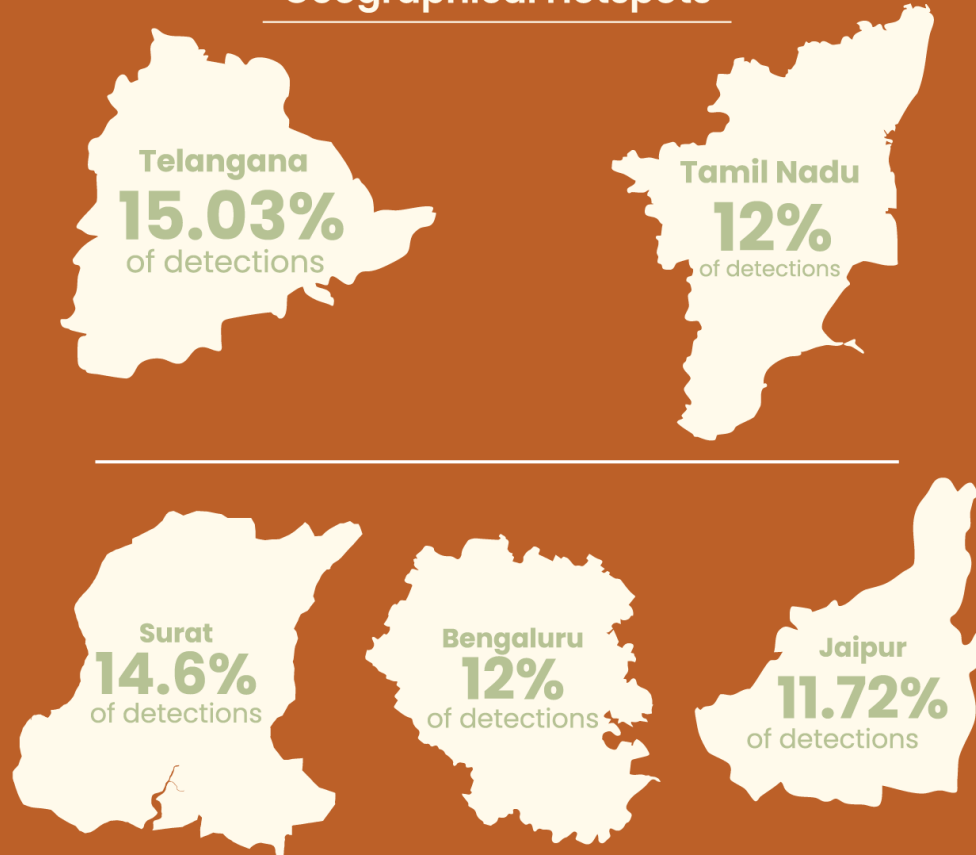


Key components of Endpoint Security



Key Highlights of India Threat Report

Geographical Hotspots



Top most targeted industries



Malware and ransomware incidents

1 Malware
incident per
40,436
detections

1 Ransomware
incident per **595**
detections

Approximately
1 million
Ransomware
detections reported
over the year



Key Highlights of India Threat Report

Bigger spectrum of
detections and endpoints

**Over
369 million**
detections recorded
across **8.44
million** endpoints

Averaging **702
detections**
per minute

Behavioral-based
detection

53.73 million
behavior-based detections,
accounted for **14.56%**
of total detection

Significant increase
from **12.5%** in 2023

~12 attacks in a
month on
personal end
user devices

Primary attack vectors

68%

of attacks originated
from Trojans and Infectors

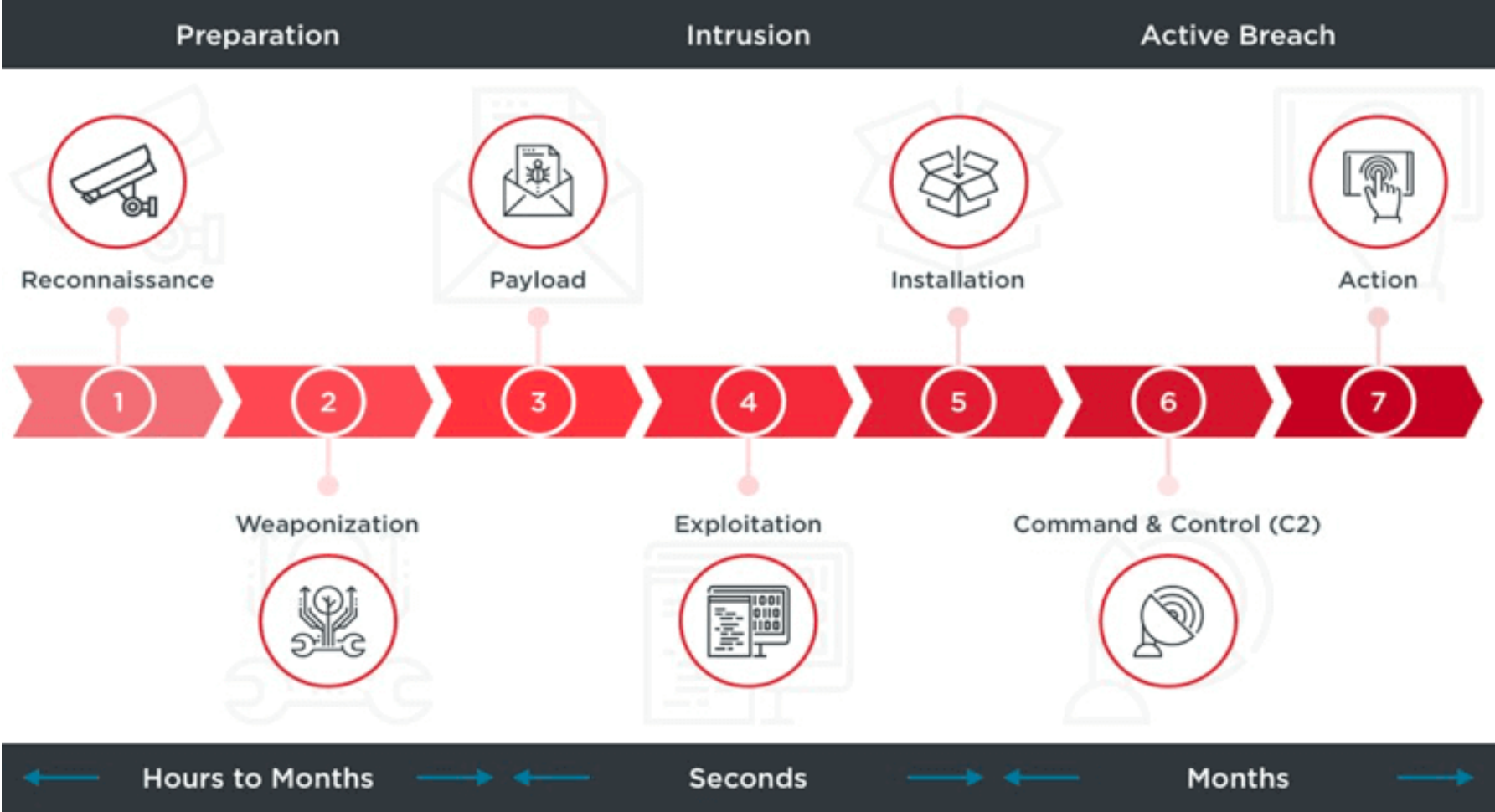
Cloud detections
contributed

62%

while 38% detections
were on-premise

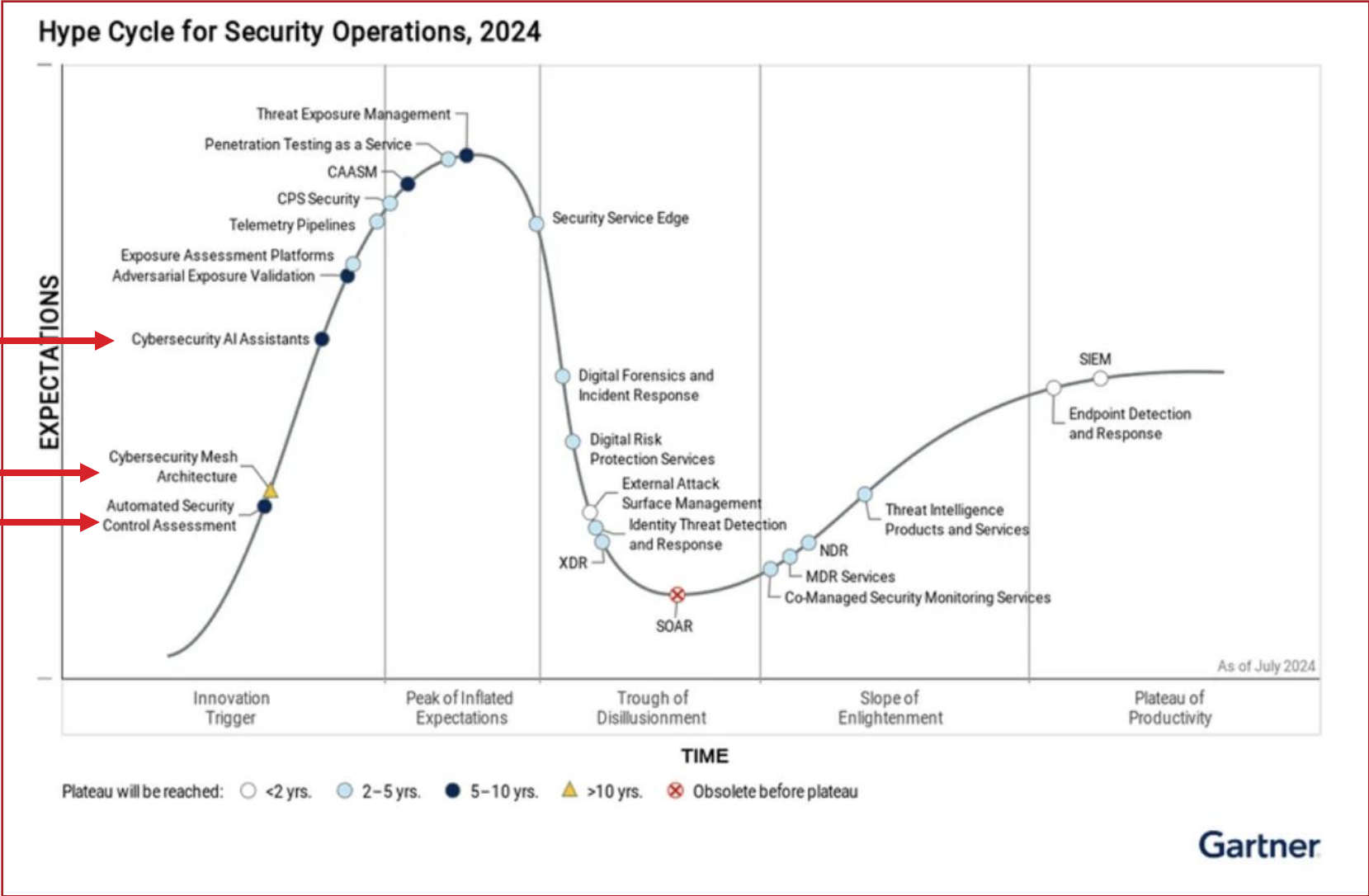


Cyber Kill Chain



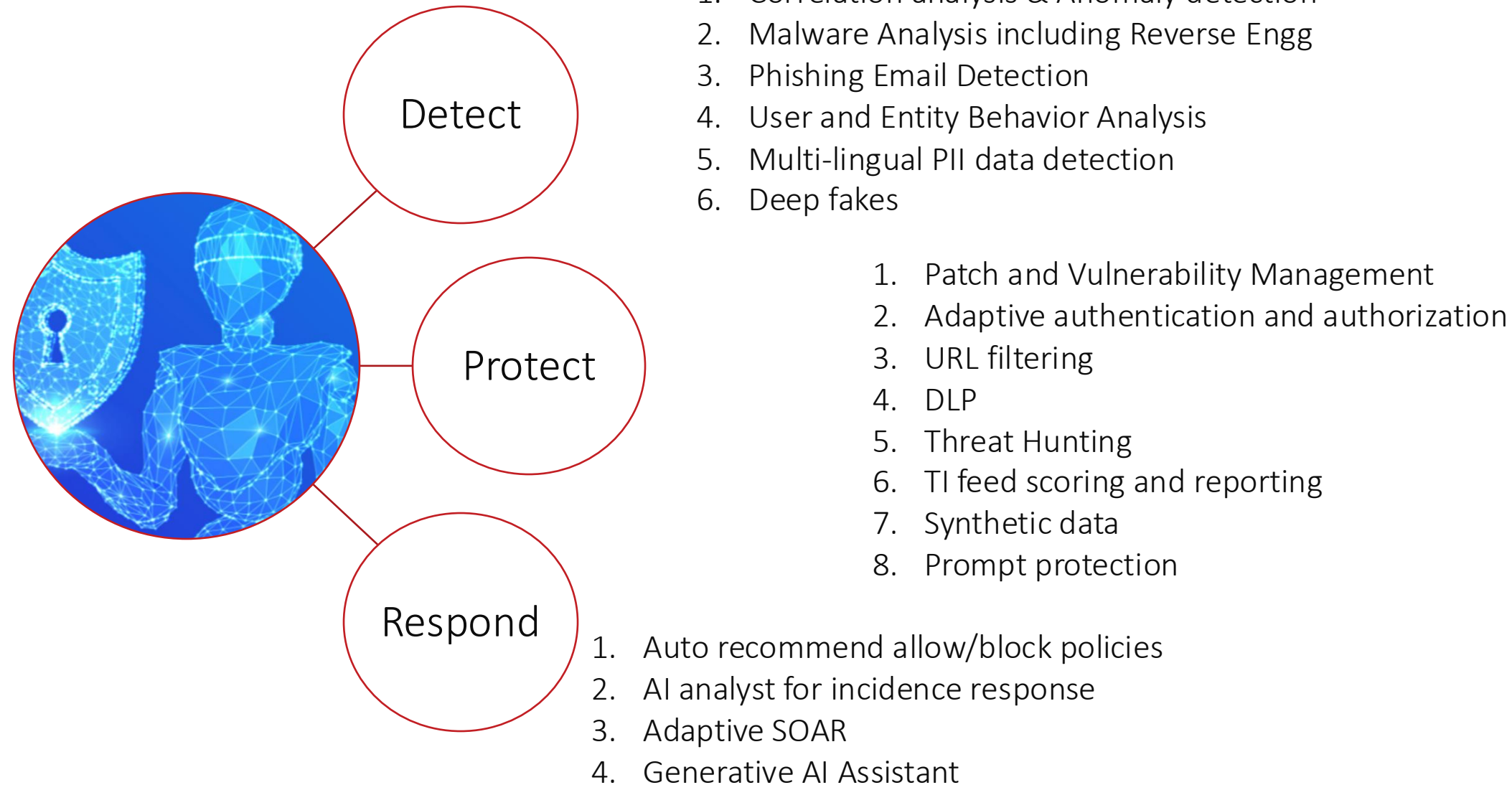


Gartner's Hype Cycle on Security Operations





Common AI/ML use cases in Cyber Security



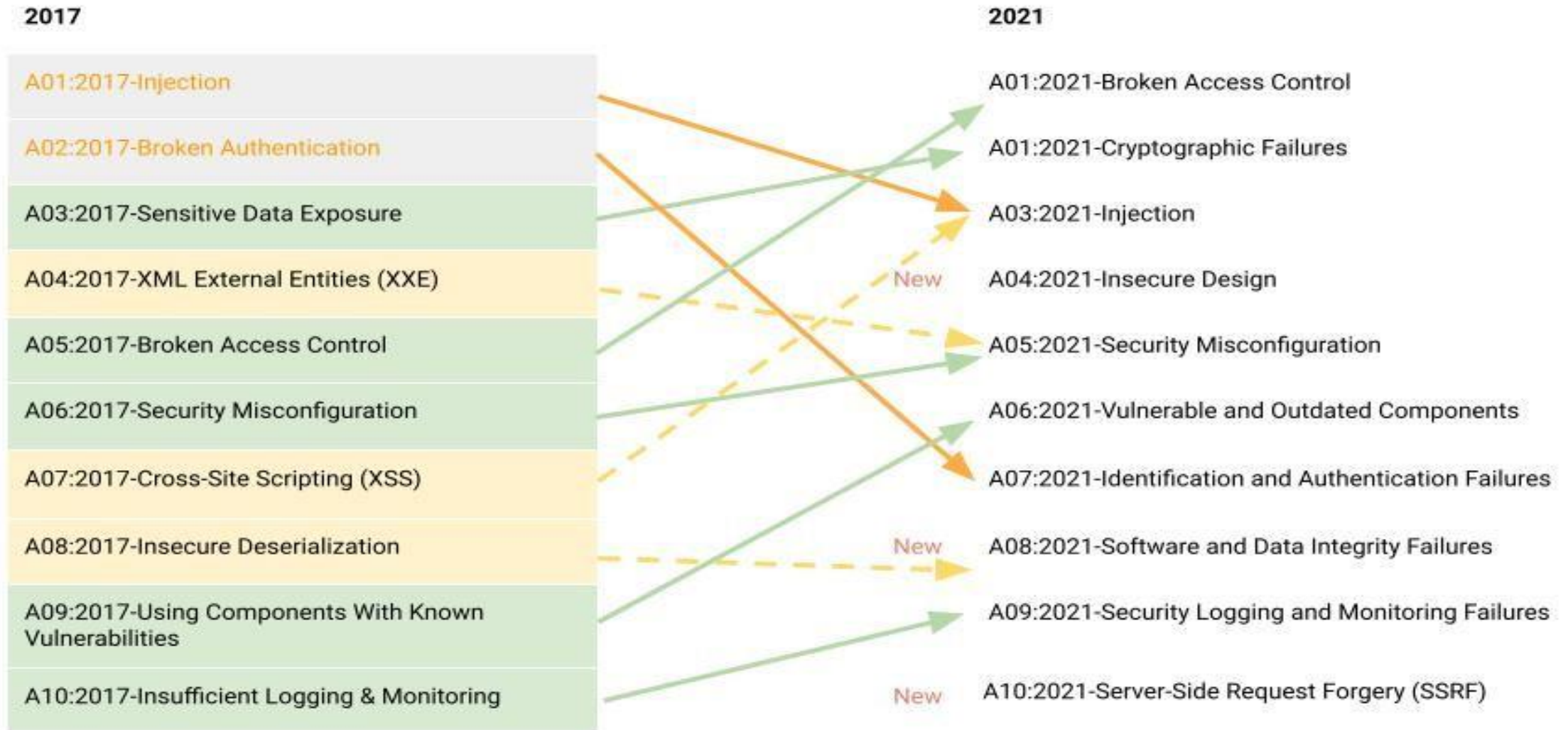


ATLAS Matrix

Reconnaissance&	Resource Development&	Initial Access&	ML Model Access	Execution&	Persistence&	Privilege Escalation&	Defense Evasion&	Credential Access&	Discovery&	Collection&	ML Attack Staging	Exfiltration&	Impact&
6 techniques	11 techniques	5 techniques	4 techniques	4 techniques	4 techniques	2 techniques	5 techniques	1 technique	6 techniques	3 techniques	4 techniques	4 techniques	7 techniques
Search for Victim's Publicly Available Research Materials	Acquire Public ML Artifacts	ML Supply Chain Compromise	AI Model Inference API Access	User Execution &	Poison Training Data	LLM Plugin Compromise	Evade ML Model	Unsecured Credentials &	Discover ML Model Ontology	ML Artifact Collection	Create Proxy ML Model	Exfiltration via ML Inference API	Evade ML Model
Search for Publicly Available Adversarial Vulnerability Analysis	Obtain Capabilities &	Valid Accounts &	ML-Enabled Product or Service	Command and Scripting Interpreter &	Backdoor ML Model	LLM Jailbreak	LLM Jailbreak		Discover ML Model Family	Data from Information Repositories &	Backdoor ML Model	Exfiltration via Cyber Means	Denial of ML Service
Search Victim-Owned Websites	Develop Capabilities &	Evade ML Model	Physical Environment Access	LLM Prompt Injection	LLM Prompt Self-Replication		LLM Trusted Output Components Manipulation		Discover ML Artifacts	Data from Local System &	Verify Attack	Extract LLM System Prompt	Spamming ML System with Chaff Data
Search Application Repositories	Acquire Infrastructure	Exploit Public-Facing Application &	Full ML Model Access	LLM Plugin Compromise	RAG Poisoning		LLM Prompt Obfuscation		Discover LLM Hallucinations		Craft Adversarial Data	LLM Data Leakage	Erode ML Model Integrity
Active Scanning &	Poison Training Data	Phishing &					False RAG Entry Injection		Discover AI Model Outputs				Cost Harvesting
Gather RAG-Indexed Targets	Establish Accounts &								Discover LLM System Information				External Harms
	Publish Poisoned Models												Erode Dataset Integrity
	Publish Hallucinated Entities												
	LLM Prompt Crafting												
	Retrieval Content Crafting												



OWASP Top 10





Thank You

Quick Heal Technologies Limited

Marvel Edge, Office No. 7010 C & D, 7th Floor, Viman Nagar, Pune - 411014, India.

Support Number: 1800-212-7377 | info@seqrите.com | www.seqrите.com